DE–225 **13**

DISTANCE EDUCATION

B.Sc. (Computer Science) EXAMINATION,
DECEMBER 2014.

DIGITAL COMPUTER FUNDAMENTALS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Convert the following

   (i) $(AB6)_{16} = ()_{10}$

   (ii) $(7574)_8 = ()_{10}$

   (iii) $(11011.10111)_2 = ()_8$

   (iv) $(426)_8 = ()_2$

   (b) Perform $(50)_{10} - (100)_{10}$ using two's complement arithmetic and verify the result.

2. (a) State and prove the Demorgan's theorems

   (b) Obtain simplified expressions in sum of products for the Boolean expression.

   $F(A,B,C,D,E) = \sum(0,1,4,5,16,17,21,25,29)$

3. (a) Explain the working principle of multiplexer and demultiplexer with diagram.

   (b) Usings maps, simplify the following expressions in four variables W, X, Y, Z.

   $m_0 + m_2 + m_4 + m_8 + m_9 + m_{10} + m_{11} + m_{12} + m_{13}$

4. (a) With diagram, explain the working of half adder and full adder circuits.

   (b) Explain the construction of ALU and its various functions.

5. (a) Explain a 3 to 8 line decoder circuit with truth table.

   (b) Explain the parallel binary adder with necessary diagram.

6. (a) With neat circuit diagram and truth table, explain the operation of JK flip-flop and how the for bidden states are avoided.

   (b) Explain the concept of serial in and serial out shift register.

7. (a) Draw the circuit of a BCD counter and using timing diagrams explain its working.

   (b) What are the different classifications of computer? Explain.

8. (a) What are the different kinds of input devices? Explain any two in detail.

   (b) What are the different generations of computers? Explain.

DE–226 **14**

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, DECEMBER 2014

DATA STRUCTURES USING 'C'

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the various operators available in 'C' with example. (10)

   (b) What are the various control statements? Explain. (10)

2. (a) Define Function. Explain the concept of passing arguments to a function. (10)

   (b) Briefly explain the various string functions in 'C'. (10)

3. (a) What is Preprocessor? Explain the various preprocessor statements with example. (10)

   (b) Explain the various library functions available in 'C'. (10)

4. (a) Define Pointer. Explain dynamic memory allocation with example. (10)

   (b) Explain array of pointers with example. (10)

5. (a) Write a 'C' program for creating student file. (10)

   (b) Explain the differences between union and structure. (10)

6. (a) Define linked list. How to insert item in a singly linked list? Explain. (10)

   (b) Explain the applications of tree. (10)

7. (a) Explain the evaluation of expression with stack. (10)

   (b) Explain the procedure for deleting a data item in a stack. (10)

8. (a) Define Array. Write a 'C' program for multiplying two matrices. (10)

   (b) What is Recursion? Explain with an example. (10)

DE–227 **15**

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, DECEMBER 2014.

DISCRETE MATHEMATICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Show that $(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$. (6)

   (b) Show that $(P \to Q) \wedge (R \to Q) \Leftrightarrow (P \vee R) \to Q$. (8)

   (c) Construct the truth table for the formula $(P \wedge Q) \vee (\neg P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$. (6)

2. (a) Show that $A \cap (B - C) = (A \cap B) - (A \cap C)$. (5)

   (b) Let a relation R be defined on the set of all real numbers by if x, y are real numbers, $xRy \Leftrightarrow x - y$ is a rational number. Show that the relation R is an equivalence relation. (10)

   (c) Give one example of a relation R on $A = \{1, 2, 3, 4\}$ which is neither symmetric nor anti symmetric. (5)

3. (a) Let $f : A \to B$ and $g : B \to C$ be functions. Prove that

   (i) if both f and g are one-to-one, $g.f$ is also one-to-one

   (ii) if both f and g are onto, $g.f$ is also onto. (10)

(b) Let $f : A \to B$ and $g : B \to C$ be both one-to-one and onto functions. Prove that $(g.f)^{-1} = f^{-1}.g^{-1}$. (10)

4. (a) Prove that for any commutative monoid $(M,*)$, the set of idempotent elements of M forms a submonoid. (6)

(b) If G is an abelian group, show that $(ab)^n = a^n.b^n$ for all $a,b \in G$ and for every integer n. (8)

(c) Prove that intersection of any two subgroups of a group G is again a subgroup of G. (6)

5. (a) State and prove Lagrange's theorem. (8)

(b) If $f : G_1 \to G_2$ is a group homomorphism, prove that its Kernel K is a normal subgroup in G_1. (6)

(c) Let G be a group and $a \in G$. Prove that the map $f : G \to G$ defined by $f(x) = axa^{-1}$ for all $x \in G$, is an isomorphism. (6)

6. (a) Prove that a simple graph with n vertices and K components can have at most $(n-K)(n-K+1)/2$ edges. (10)

(b) Prove that in a simple digraph $G = \langle V, E \rangle$, every node of the digraph lies in exactly one strong component. (10)

7. (a) Define planar graph. Prove that K_5 is not planar. (8)

(b) Prove that a graph is a tree if and only if it is minimally connected. (6)

(c) Prove that a tree with n vertices has $(n-1)$ edges. (6)

8. (a) Show that $A \cup (B - C) = (A \cup B) - (C - A)$. (5)

(b) Prove that the relation "congruence modulo m" over the set of positive integer is an equivalence relation. (10)

(c) Let $X = \{1, 2, 3, 4\}$ and a function $f : X \to X$ be given by $f = \{(1,2), (2,3), (3,4), (4,1)\}$. Form the composition f^2, f^3 and f^4. (5)

DE–228 **23**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

OBJECT ORIENTED PROGRAMMING AND C++

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) What are the basic concepts of object oriented programming? Explain with an example.

   (b) Explain the various data types supported by C++.

2. (a) What are the different operators used in C++? Explain with suitable examples.

   (b) Explain the use of Cin and Cout with a sample program.

3. (a) Write a C++ program to find *ncr* using function factorial and treat this function as inline.

   $$\left[ncr = \frac{n!}{r!(n-r)!} \right].$$

   (b) Explain the manipulators and their use.

4. (a) With the help of a suitable example show how a friend function can act as a bridge between two classes.

   (b) How do you use static data and member functions is C++? Give an example.

5. (a) Write a C++ program to illustrate the use of THIS pointer.

   (b) Explain the uses of new and delete operators with suitable examples.

6. (a) Write a C++ program to multiply any two order of matrix using pointers.

   (b) Explain the run time polymorphism with example.

7. (a) Define a class complex. Using this class, overload '+' operator to add two complex numbers.

   (b) Explain the concept of function overloading in C++ with a programming example.

8. (a) Explain in detail the various forms of inheritance.

   (b) What is virtual base class? Explain this concept with C++ program.

DE–229 **24**

DISTANCE EDUCATION

B.Sc.(C.S.)/B.Sc. C.S. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

SCIENTIFIC COMPUTING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Solve the following equations by Gauss-Seidal method.

$4x + 2y + z = 14$; $x + 5y - z = 10$; $x + y + 8z = 20$.

(b) Solve the system of equations by Guass-elimination method.

$10x - 2y + 3z = 23$; $2x + 10y - 5z = -33$;

$3x - 4y + 10z = 41$.

2. (a) Evaluate $f(4)$ using Newton's backward formula to the following data :

x:	0	1	2	3
$f(x)$	1	2	1	10

(b) By dividing the range into ten equal parts, find the value of $\int_0^{\pi} \sin x dx$.

3. (a) Find the Arithmetic mean for the frequency distribution

Marks:	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of students:	5	9	14	20	25	15	8	4

(b) Find the mean deviation about the mean for the following data:

x:	10	11	12	13	14
f:	3	12	18	12	3

4. (a) Find the standard deviation of the salaries for the following data:

Salary (Rs.):	75	80	85	90	95	100
No. of persons:	3	7	18	12	6	4

(b) Fit a straight line to the following data:

x:	7	6	7	8	8	8	9	9	10
y:	5	5	4	3	4	5	4	3	3

5. (a) If A and B are any two events, Prove that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

(b) A company has three machines M_1, M_2, M_3, which produces 20%, 30% and 50% of products respectively. Their respective defective percentages are 7, 3 and 5. From these products one is chosen and inspected. It is defective, what is the probability that it has been made by machine M_3?

6. (a) A random variable x has the following probability function:

x:	0	1	2	3	4	5	6	7
$p(x)$:	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Find

(i) k

(ii) $P(x < 6)$

(iii) $P(x \geq 6)$

(iv) $P(0 < x < 5)$

(b) Find the characteristic function of

$$f(x)=\begin{cases}\dfrac{1}{b-a}, & a<x<b\\ 0 & \text{otherwise.}\end{cases}$$

7. (a) Two random samples gave the following results:

$n_1 = 10,\ \sum(x_i-\bar{x})^2 = 90$

$n_2 = 12,\ \sum(y_i-\bar{y})^2 = 108$.

Test whether the samples came from the populations with same variance.

(b) In 120 throws of a single die, the following distribution of faces was observed.

Face:	1	2	3	4	5	6
Frequency:	30	25	18	10	22	15

Using χ^2–test, can you say that the die is biased?

8. Perform two-way ANOVA for the data given below:

	Treatment			
Plots of Land	A	B	C	D
I	38	40	41	39
II	45	42	49	36
III	40	38	42	42

Use coding method, subtract 40 from the given numbers.

DE–230 **25**

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

APPLICATION PROGRAMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain in detail about any two Icons available in Windows Desktop. (10)
   (b) Write a detailed note on various Dialog Boxes on Windows with Relevant examples. (10)

2. (a) Discuss in Detail about the various Formatting Options available in Microsoft Word. (10)
   (b) Write a note on Print Preview and Page Numbering Options in Microsoft word. (10)

3. (a) Explain how to Insert Headers and Footers and change Line Spacing in Microsoft Word. (10)
   (b) Explain in Detail the uses of Formula Bar in Microsoft Excel. (10)

4. (a) Write a Detailed note on Conditional Formatting in Microsoft Excel. (10)
   (b) Discuss in Detail about the various Steps involved in Creating a Chart in Microsoft Excel. (10)

5. (a) Explain how to build a Database in Microsoft Excel and Sort it. (10)
   (b) Write a Detailed note on Creating and Editing a Presentation in Microsoft Power Point. (10)

6. (a) Explain in detail about how to perform Grouping and Ungrouping in Microsoft Power Point. (10)

   (b) Write a note on using Design Templates and adding Speaker notes in Microsoft Power Point. (10)

7. (a) Discuss briefly about Creating Handouts in Microsoft PowerPoint. (10)

   (b) Explain in Detail about how to create a new Database and perform various operations in it. (10)

8. (a) Write a Detailed note on Adjusting the Field type and size in Microsoft Access. (10)

   (b) Discuss on Auto form Feature of Microsoft Access in detail. (10)

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DE–231 **31**

DISTANCE EDUCATION

B.Sc.(CS)/B.Sc. (Computer Science)(Lateral) DEGREE EXAMINATION, DECEMBER 2014.

COMPUTER GRAPHICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain Bresenham's algorithm for line drawing. (10)

   (b) Explain circle generation algorithm with example. (10)

2. (a) Explain the various types of two dimensional transformation with matrix representations. (10)

   (b) Explain the windows and view ports with example. (10)

3. (a) Explain three dimensional rotation with matrix representation. (10)

   (b) Discuss on three dimensional composite transformation. (10)

4. (a) Discuss on viewing transformation. (10)

   (b) Explain the differences between point and line clipping. (10)

5. (a) Define projection. Explain the various types of projection. (10)

   (b) Explain Cohen Sutherland algorithm for clipping. (10)

6. (a) Explain in detail about command language. (10)

   (b) Discuss the components of user interface. (10)

7. (a) Explain the z-buffer algorithm. (10)

   (b) Explain clipping process in a three dimensional plane. (10)

8. (a) Explain the various output devices with example. (10)

   (b) Write short note on command language. (10)

DE–232 **32**

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (Computer Science) (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

VISUAL PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. (5 × 20 = 100)

1. (a) Write about the hardware and software requirements for windows progamming. (10)

   (b) Write about selecting and resizing windows. (10)

2. (a) Write briefly about the features available in print manager. (10)

   (b) Write briefly about PIF editor. (10)

3. (a) Explain with example, how the menu editor performs the following

      (i) Add new menus

      (ii) Delete old menus (10)

   (b) Explain the inputbox and messagebox functions with example. (10)

4. (a) With example, explain the different forms of If statement. (10)

   (b) Write about the properties of command button and text box. (10)

5. (a) Explain in detail about procedures in VB. (10)

   (b) Explain control arrays with example. (10)

6. (a) Write briefly about windows help system. (10)

   (b) Write about object linking and embedding. (10)

7. (a) Explain the following

      (i) For....Next

      (ii) While....wend (10)

   (b) With a sample program, explain about event-driven programming. (10)

8. (a) Write a VB program to convert a decimal number to binary number. (10)

   (b) Explain MDI forms with example. (10)

DE–233 **33**

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. Computer Science (Lateral) DEGREE EXAMINATION, DECEMBER 2014

OPERATING SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is an operating system? Explain in detail about Timesharing and Multiprocessing systems. (10)

   (b) Explain in detail about the various category of system programs in operating system. (10)

2. (a) Discuss in detail about the Process Control Block (PCB). Explain with relevant diagram. (10)

   (b) Write a detailed note on synchronization and buffering. (10)

3. (a) Explain shortest job first scheduling algorithm. (10)

   (b) Discuss in detail about Round Robin Scheduling. (10)

4. (a) Write a detailed note on Device controllers. (10)

   (b) What is Deadlock? Explain how to detect and correct it. (10)

5. (a) Discuss in detail about multiprogramming with fixed and variable partitions. (10)

   (b) Explain in detail about linked lists and buddy system. (10)

6. (a) Discuss about multiprogramming with out swapping. (10)

   (b) Write a detailed note on paging and segmentation. (10)

7. (a) What is a file? Explain in detail about the various attributes which are common for all file systems.(10)

   (b) Explain any two methods of accessing files in detail. (10)

8. (a) What is a Directory? Explain about Directory Structures. (10)

   (b) Write as detailed note on file protection mechanisms. (10)

DE–234 **34**

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

INTERNET PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Categorize the five basic topologies and explain it. (10)

   (b) Discuss the four layers of the TCP/IP protocol Suite. (10)

2. (a) Describe the design issues of layers of ISO OSI Reference Model. (10)

   (b) What is FTP and how does it work? (10)

3. (a) What are the advantages and disadvantages of E-mail? (10)

   (b) State the types of URL and explain them. (10)

4. (a) What is the purpose of HTTP? Discuss any three HTTP commands. (10)

   (b) Discuss the various services provided by an Internet. (10)

5. (a) What is E-Commerce? Describe the applications of E-Commerce. (10)

   (b) Describe the structure of Java Program. (10)

6. (a) What are the classifications of operators in Java? Explain with examples. (10)

   (b) Explain the fundamental characteristics of Object Oriented Programming. (10)

7. (a) Describe the different stages in the life cycle of an applet. (10)

   (b) Write a Java program that creates an abstract class called dimension creates two subclasses, rectangle and triangle. Include, appropriate methods for both the subclasses that calculate and display the area of the rectangle and triangle. (10)

8. (a) How does Java handle strings? Explain with example. (10)

   (b) Describe the different forms of inheritance with examples. (10)

DE–235 **35**

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

RDBMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What are the advantages and disadvantages of Database Management System? Describe. (10)

   (b) Write a short note on Schema and Subschema. (10)

2. (a) Discuss the typical user friendly interfaces and the types of users who use each. (10)

   (b) Explain the concepts of relational data model. Also discuss its advantages and disadvantages. (10)

3. (a) Define the terms entity, attribute, role and relationship between the entities, and give examples for each of them. (10)

   (b) Differentiate between DDL and DML. (10)

4. (a) What are the key features of Oracle? (10)

   (b) What is NULL? Give an example to illustrate testing for NULL in SQL. (10)

5. (a) What is Oracle Process? Explain any four processes started by Oracle. (10)

   (b) In Oracle what is system global area and how is it organized? (10)

6. (a) Differentiate between WHERE and HAVING clause in SQL. (10)

   (b) What is a sub query? Explain with an example. (10)

7. (a) What is an INDEX as defined in ORACLE? Write the syntax of creating an INDEX. Create an index for the table Client, field CLIENT_NO of Q. (10)

   (b) What are the features of embedded SQL? Explain. (10)

8. (a) Describe DROP TABLE command of SQL with both the options — CASCADE and RESTRICT. (10)

   (b) Describe the GRANT function and explain, how it relates to security? What types of privileges may be granted? How are they revoked? (10)

DE–8820 **16**

DISTANCE EDUCATION

B.Sc. DEGREE EXAMINATION, DECEMBER 2014.

Computer Science

COMPUTER LAB I (C AND DATA STRUCTURES LAB)

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the students by LOT system.

1. (a) A man is paid at the hourly rate of Rs.30 per hour for the first 50 hours worked. Thereafter, overtime is paid at 2 times the hourly rate for the next 25 hours and 3 times the hourly rate for further hours worked per week. Calculate and print his gross weekly wage.

   (b) Write a C program to implement push and pop operations on stack.

---------- Cut here ----------

2. (a) Write a C program that receives the data such as age and name of person to check the eligibility for voting. Take the condition that if a person is more than 18 years old he is eligible to vote. Otherwise display the number of years he has to wait for voting.

   (b) Write a C program to evaluate the given mathematical expression using stack.

3. (a) There are 10 students in a class. Their names and marks in three different subjects are given. If a student takes more than 40 marks in each subject, then he is declared 'PASS'. Otherwise 'FAIL'. Write a C program to do the above.

   (b) Write a C program to implement insert and delete operations on linked list.

---------- Cut here ----------

4. (a) Write a C program to arrange the given set of numbers in descending order.

   (b) Write a C program to implement insert and delete operations on queue using array concept.

---------- Cut here ----------

5. (a) Write a C program to find the maximum and minimum in a given set of numbers.

   (b) Write a C program for linked list implementation of queue operations.

---------- Cut here ----------

6. (a) Write a C program to concatenate two given strings and find the length of the concatenated string.

   (b) Write a menu driven program in C to find an element using linear and binary search.

DE–8821 **26**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

LAB II — C++ PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by LOT system.

Each sub-division carries 50 marks.

1. (a) Write a temperature conversion program that gives the user the option for converting Fahrenheit to celcius or celcius to Fahrenheit. Then carry out the conversion. Use floating point numbers.

   (b) Write a program that emulates the DOS copy command. The program should give error message if the destination file already exists.

---------- Cut here ----------

2. (a) Write a menu driven C++ program to add and subtract given two matrices of order $m \times n$ defined in a class. Use operator overloading concept.

   (b) Create a class Int to overload all five integer arithmetic operators (+, –, *, / and %) so that they operate on objects of type Int. If the result of any such arithmetic operation exceeds the normal range of integer from –32768 to 32767 then print a warning and terminate the program. Write a program to test this class.

3. (a) Create a class DONOR that contains donor number, donor name, age, address, sex, and blood group.

Write a menu driven C++ program to display the number, name and address of the donors for the following categories.

(i) Blood donors having the blood group O+.

(ii) Blood donors in the age group between 16 to 25.

(iii) Female donors having blood group A is the age between 19 and 24.

(b) Write a program using friend function frifunc ( ) which can act on the classes alpha and beta. Using constructors fix the values for alpha and beta.

---------- Cut here ----------

4. (a) Using dynamic constructors write a C++ program to concatenate two given strings.

(b) Write a C++ program which will accept a string of 10 characters from the keyboard and count the occurences of each of the five vowels in the string. The output should be in a (tapped) format similar to this example :

a	e	i	o	u
2	3	0	3	2

---------- Cut here ----------

5. (a) Write a program to process students marks with the help of classes. The class has private variables for name, mark 1, mark 2, mark 3. It has two member functions : getdata ( ) – to get input ; result ( ) – to print the results. All subjects mark must be greater than or equal to 50 for pass, otherwise fail.

(b) Write a program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

6. (a) Create a class employee that contains a employee number, employee name and address. Write a menu driven C++ program to get the *n* number of employee details and display all details in employee name wise sorted order.

(b) Write a C++ program to design a calculator with four arithmetic operations such as addition, subtraction, multiplication and division.

DE–8822 **27**

DISTANCE EDUCATION

B.Sc.(CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

COMPUTER LAB – III

(APPLICATION PROGRAMS LAB)

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select TWO questions One from Section-A and another from Section-B.

The questions are given to the candidates by LOT system.

SECTION A – (50 marks)

1. Prepare an Invitation for a function to be conducted in your institution use different text orientation and pictures to make it attractive.

---------- Cut here ----------

2. Create a work sheet with the following columns.

   Salesman number, Salesman Name, City, Product name, Sale Amount.

   Add three records for five different salesman who have carried out sales of different products in different cities. Find the following :

   (a) City wise total sales

   (b) Salesman wise total sales

   (c) Product wise total sales

   (d) Grand total sales.

3. Draw Line, Bar, PIE charts for the data given below :

ABC Company Limited

Year	Sales	Cost	Profit
1991	1000	400	600
1992	1500	600	1610
1993	1600	650	1610
1994	1200	650	1020
1995	1250	650	950
1996	1200	650	800
1997	1400	500	1000
1998	2000	500	1200
1999	2050	500	1100
2000	2100	550	850

Provide titles, legends, grids and data labels.

---------- Cut here ----------

SECTION B – (50 marks)

4. Create employee table with the fields EMPNO, NAME, AGE, SEX, STREET, CITY, PIN, SALARY.

(a) Add the data for 25 employees

(b) Write a query to display all the female employees whose salary is between 750 and 2500 and living in city " ERODE".

(c) Write a query to display all the male employees whose ages are in the range of 40-45

(d) Show all the records in the table for the city "SALES" by filtering.

5. Create STUDENT table with the following fields REGNO, NAME, SEX, BLOOD GROUP, HEIGHT, WEIGHT, STREET, CITY and PIN.

   (a) Write a query to display all the male students whose blood group is O+ve

   (b) Write a query to display all the girl students whose height is between 130 to 140 and weight is between 40 to 50.

---------- Cut here ----------

6. Create inventory table with fields ITEM No, NAME, QUANTITY ON HAND, REORDER LEVEL.

   (a) Create a form in custom format.

   (b) Create a query REORDER to show all the items.

   (c) Create a macro that executes REORDER query automatically.

DE–8823 **36**

DISTANCE EDUCATION

B.Sc. (Computer Science/B.Sc. (CS) Lateral) DEGREE EXAMINATION, DECEMBER 2014.

COMPUTER LAB IV — VISUAL BASIC AND ORACLE LAB

(2007 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and it is given to the candidate by LOT system.

1. (a) Write and test a VB program to print ten address labels in a two-column format for the address :

The Registrar

University of Madras

Chennai – 600006.

(b) Create a table MARK with the following fields :

REGNO, NAME, TAMIL, ENGLISH, MATHS, SCIENCE

(i) Add 6 records

(ii) Show data in fields REGNO, NAME, TAMIL and MATHS alone on the screen

(iii) Show the records those who scored 100 in MATHS

(iv) Show all fields on the screen.

2. (a) Write a VB program to compute the surface area and volume of a sphere given the radius. Use option buttons and INPUT boxes.

(b) Create a table RESULT with the following fields :

REGNO, NAME, MARK1, MARK2, MARK3, TOTAL, RESULT.

(i) Add 10 records (except TOTAL and RESULT)

(ii) Fill TOTAL = MARK1 + MARK2 + MARK3

(iii) Fill the RESULT field with 'PASS' if total ≥ 150 otherwise 'FAIL'.

(iv) Show all fields on the screen.

---------- Cut here ----------

3. (a) Write a VB program to compute and print either the sum of odd numbers or even numbers at the user's choice using label, text and option buttons.

(b) Create a table BIODATA with the following fields :

NAME, AGE, SEX, DEGREE, CITY.

(i) Add 10 records and display all male records.

(ii) Delete all B.Sc. degree holders and then display

(iii) Display all records having age < 30 and CITY is 'CHENNAI'

(iv) Remove all records whose age is >20.

4. (a) Write a VB program that allows the user choice among four arithmetic operations of addition, subtraction, multiplication and division with two given numbers.

(b) Create a table PERSONAL with the following fields :

SNO, NAME, CITY, PIN, PHONE

(i) Add 6 records and display the contents

(ii) Display all names without duplicates

(iii) Display all names and length of names

(iv) Display name, city in which only beginning portion of name is known.

---------- Cut here ----------

5. (a) Write a VB program to display the day of the given date and convert upper case from lower case of the given word.

(b) Create a table SALES with the following fields :

SALENO, SALENAME, SALEAMT, COMM, DATE

(i) Add 5 records

(ii) Replace COMM with the following conditions :

If SALEAMT < 1000; No commission

If SALEAMT ≥ 1000 and < 5000; Commission = 5% of sales amount.

If SALEAMT ≥ 5000, Commission = 10% of sales amount

(iii) Display all records

(iv) Display the records having commission > 1000.

DE–8824 **37**

DISTANCE EDUCATION

B.Sc. (Computer Science /B.Sc. (CS) lateral) DEGREE EXAMINATION, DECEMBER 2014.

COMPUTER LAB – V (JAVA PROGRAMMING LAB)

(2007 onwards)

Time : Three hours Maximum : 100 marks

One question should be selected by the examiner for each candidate and given to them by LOT system.

Each subdivision carries 50 marks.

1. (a) Write a java program to illustrate the implementation of abstract class.

   (b) Write a java program to count the number of characters, lines and words in a text file.

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2. (a) Write a java program for sorting a given list of names in ascending order using command line arguments.

   (b) Write a java program to read and write the data using random access file.

---------- Cut here ----------

3. (a) Write a java program to multiply the given two matrices.

   (b) Write a java program to create a package.

4. (a) Write a program to overload the + operator to provide string addition.

   (b) Write a java program to create an applet and draw the various polygons.

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5. (a) Write a program to create multiple threads in java.

   (b) Write a java program to display the contents of file along with a line number before each line.

---------- Cut here ----------

6. (a) Write a java program to handle mouse and key board events.

   (b) Write a java program which illustrate the use of files and streams.

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7. (a) Write a java program to count the number of words, characters and lines in a text.

   (b) Write a java program to implement producer/ consumer problem using synchronization.

---------- Cut here ----------

8. (a) Write a java program to implement the multiple inheritance using interfaces.

   (b) Write a java program to create and manipulate labels, lists, text fields, text areas and panels.